Urban Light Pollution and Avian Spatial Distribution: Evidence and Policy Recommendations for European Cities

Prepared by Group B: Mia Corduneanu-Huci, Shiwam K.C., Thanika Haltrich — Data Science Project UGST4092 — April 12, 2026

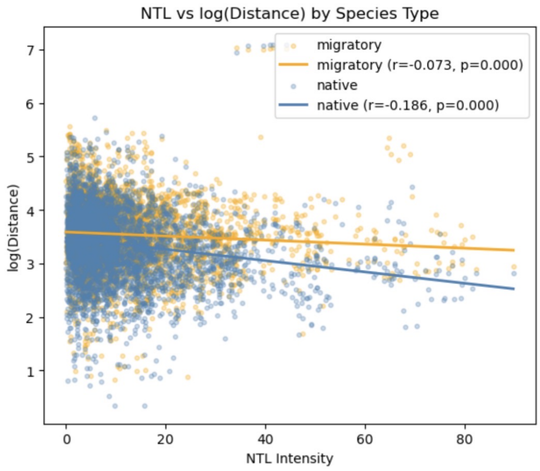
Executive Summary. Every autumn, hundreds of millions of birds cross Europe at night, navigating by stars and the Earth's magnetic field, a system refined over millions of years. However, human activity is changing their behavior. It is no secret that we affect our environment - we produce air pollution, we expand our cities, and we use artificial lights at night. We have the idea in the back of our minds that our actions have consequences, *but how much?* Our study *Quantifying Anthropogenic Influence on Spatial Avian Distribution* finds that human activity - particularly the use of artificial nightlights - truly has an effect on avians. We are directly responsible for drawing birds toward bright urban areas; where they become disoriented, stuck near city centres, and are killed. This policy brief aims to showcase the evidence found in this study and encourage policy makers to address the issue of regulating urban nightlight usage throughout the year.

Key Findings

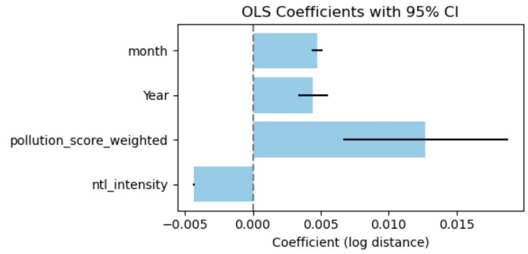
European cities are among the most light-polluted areas on the continent. Their ecological footprint on surrounding wildlife continues to be substantial. Birds, among the most ecologically sensitive indicators of environmental change, are increasingly being observed closer to urban centres across Europe during the years 2017 - 2022. Our analysis of over 4 million bird observations from the bird sighting platform eBird during this time frame, matched to satellite nightlight intensity images from NASA, and air quality data finds a consistent and statistically significant negative relationship between nightlight intensity and bird observation distance (Figure 1b). This means that the brighter a city, the closer birds are recorded to its centre. This effect holds across all six years of the study period and holds for both native and migratory birds. However, residents are approximately 2.5 times more sensitive to urban light than migrants (Figure 1a). This difference is ecologically meaningful, as native birds are constantly influenced by artificial light while migratory species pass through cities and urban areas only temporarily.

These findings carry a clear policy implication. The strongest signal in our data is not pollution, not season, and not year, but it is **nightlight intensity**. It explains variance in bird spatial distribution, which also supports previous research findings [2]. We had also conducted experiments before, during, and after the COVID-19 period, because this is when human activity decreased due to lockdowns. During the pandemic, native birds moved toward cities at roughly 2.5 times the rate of migratory birds, suggesting that it is not just light that shapes where birds can go. Urban light pollution and air quality are also environmental factors that affect the spatial distribution of birds. All three of these factors should be acknowledged by European policymakers. New policies regarding them should be enforced to protect avian biodiversity across the continent.

Summary of impact: Figure 1a confirms the core finding — higher nightlight intensity is significantly associated with birds being observed closer to urban centres, with native species showing nearly $2.5\times$ *greater sensitivity than migrants. Figure 1b demonstrates that NTL intensity is the dominant negative predictor of bird-to-city distance across the OLS model, outweighing seasonal and temporal controls.*



(a) NTL vs log(Distance) by Species Type — migratory ($r = -0.073$, $p = 0.000$) and native ($r = -0.186$, $p = 0.000$), showing native species are significantly more sensitive to nightlight intensity.



(b) OLS Coefficients with 95% CI — NTL intensity shows a significant negative effect on bird-to-city distance; `pollution_score_weighted` has the largest positive coefficient.

Policy Options & Recommendation

Based on our findings, we propose the enforcement of the following policies:

Option 1. 'Lights Out' Migration Windows. Incentivize the dimming of non-essential commercial lighting during peak migration periods (April–May and August–October). An example is Chicago's Lights Out programme, which operates on exactly this principle. This can be used as a blue print for European cities, particularly those with high average radiance like Zaragoza, Athens, and London.

Option 2. Urban Spectral Standards. Not all artificial light is equally harmful. Only Short-wavelength (blue-white) LED lighting is disorienting for nocturnal migranting birds. EU member states could update outdoor lighting standards to require amber or narrow-spectrum LEDs in zones where migrating birds cross. This policy is cost efficient and technically straightforward.

Option 3. Green Buffer Zoning. Urban planning regulations could designate special buffer areas that such as parks, wetlands, and other places where birds commonly go. New commercial and residential developments within these zones would face stricter light emission limits.

Implementation & Next Steps

Effective implementation requires coordination between the EU in both national and municipal levels. 'Lights Out' policies can begin in the municipal levels during peak migration period. This can be supported by incentives and awareness campaigns. Urban spectral LED standards should be introduced through EU and national regulations, where public lighting leads the transition. Green buffer zoning can be incorporated into spatial planning by mapping key habitats and applying stricter lighting limits to new developments.

Next steps include collaborating with institutes to obtain a database of migratory routes and light pollution, launching pilot projects in high-risk cities, and aligning regulations across member states. Continuous monitoring both birds and public opinion is crucial to keep track of policies and refine them accordingly over time.

References

[1] A. Fraenkel, "The growing effects of light pollution on migratory birds," *UN Chronicle*, Oct. 2022.

[2] City of Chicago, "Lights Out Chicago," City of Chicago Environmental Programmes, [Online]. Available: https://www.chicago.gov/city/en/progs/env/lights_out_chicago.html.